aTableau

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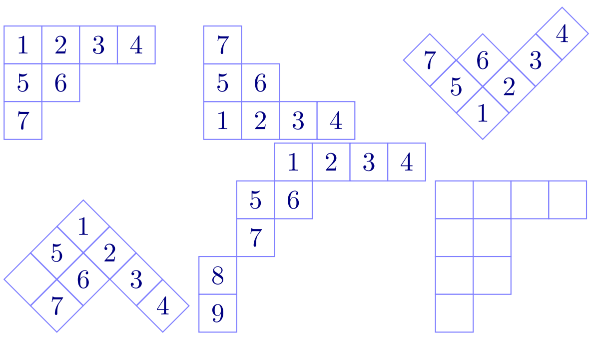
Version 2.0

Contents

Young diagrams, tableaux and abacuses are common objects used in algebraic combinatorics and representation theory. Partly because of their ubiquity, there are several different conventions in use and authors often want to decorate these objects in weird and wonderful ways.

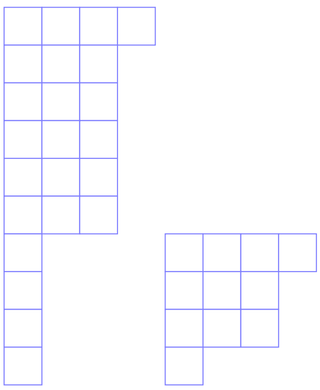
This package provides highly customisable ways of drawing combinatorial objects. The aim to to make to provide commands that are easy to use and understand and that, at the same time, give a straightforward way to further embellish the diagrams using a key-value interface. All of the common styles and conventions are supported. In addition, the commands either be used by themselves in your document or as part of a `tikzpicture` environment, when more advanced formatting is required.

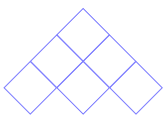


```
\Tableau{1234,56,7}
\Tableau[french]{1234,56,7}
\Tableau[ukraine]{1234,56,7}
\Tableau[australian]{1234,56,.7}
\Tableau[skew={2,1^2}]{1234,56,7,8,9}
\Tableau{....,..,..,.}
```

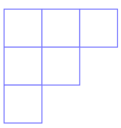
There are several other LaTeX packages available for drawing tableaux, Young diagrams etc.

1. Young diagrams

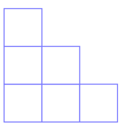




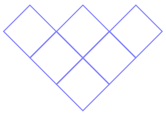
```
\Diagram[australian]{3,2,1}
```



```
\Diagram[english]{3,2,1}
```

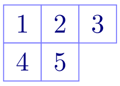


```
\Diagram[french]{3,2,1}
```

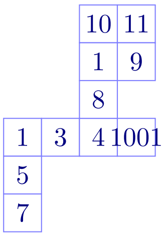


```
\Diagram[ukraine]{3,2,1}
```

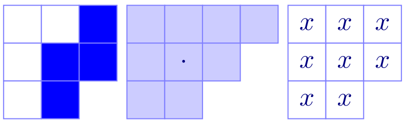
2. Tableaux



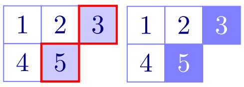

```
\Tableau{123,45}
```



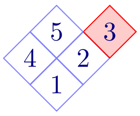

```
\begin{tikzpicture}
  \node at (1,1) {$\bullet$};
  \node at (2,2.5) {$\bullet$};
  \Tableau(1,1) {134{1001},5,7}
  \Tableau(2,2.5) {{10}{11},19,8}
\end{tikzpicture}
```



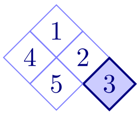
```
\Tableau[starstyle={fill=blue}]{..*., .*.*., .*.}
\Tableau[dotstyle={fill=blue!20}]{.{.}..,
.[atableau/dotstyle]\cdot ., ..}
\Tableau[dot=x]{..., ..., ..}
```



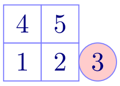

```
\tikzset{R/.style={fill=blue!20,draw=red,thick}}
\Tableau{12[R]3,4[R]5}
\Tableau[starstyle={fill=blue!50,text=white}]{12*3,4*5}
```



```
\tikzset{R/.style={fill=red!20,draw=red}}
\Tableau[ukraine]{12[R]3,45}
```

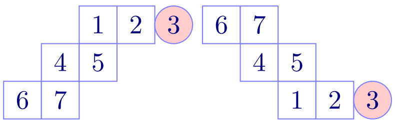


```
\tikzset{R/.style={fill=blue!20,draw=NavyBlue, thick}}
\Tableau[Australian]{12[{R, thick}]3,45}
```



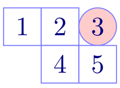

```
\tikzset{R/.style={fill=red!20,circle, radius=0.2}}
\Tableau[french]{12[R]3,45}
```

3. Skew tableau

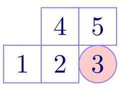



```
\tikzset{R/.style={fill=red!20,circle}}
\SkewTableau{2,1}{12[R]3,45,67}
\SkewTableau[french]{2,1}{12[R]3,45,67}
```

4. Shifted tableau




```
\tikzset{R/.style={fill=red!20,circle}}
\ShiftedTableau{12[R]3,45}
```

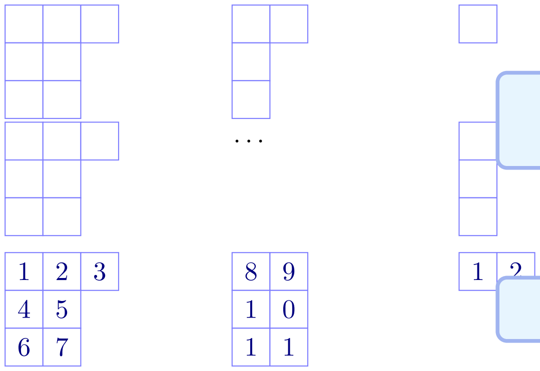



```
\tikzset{R/.style={fill=red!20,circle}}
\ShiftedTableau[french]{12[R]3,45}
```

5. Tabloids

6. Ribbon tableaux

7. Multidiagrams and multitableaux




```
\Multidiagram{3,2^2|2,1^2|1}
\Multidiagram{3,2^2|...|2,1^2}
```

```
\Multitableau{123,45,67|89,{10},{11}|{12}}
```

8. Abacuses

9. Dynkin quivers

10. Braid diagrams

11. KLRW diagrams

12. The code

Every command provided by aTableau uses TikZ, with the bulk of the code in the package being written in LaTeX3. Most of the tableau commands are special cases of the `\Tableau` command. Here is a dictionary:

Command	Equivalent `\Tableau` command
`\ShiftedTableau{...}`	`\Tableau[shifted]{...}`
`\SkewTableau{$\mu$}{...}`	`\Tableau[skew=$\mu$]{...}`
`\Tabloid{...}`	`\Tableau[tabloid]{...}`

The `\Diagram` command also invokes the `\Tableau` command, however, this requires some involved internal trickery. For example, `\Diagram{3,2^2}` is the same as `\Tableau{...,..,..}`.